

Practice Problems for Hypothesis Statements Solutions

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There will be two statistics that you will be dealing with. One of them is the population mean which we denote with μ (read as mu, with a long “u”). The other is the population proportion which we will denote with π (common is p but that is inconsistent with the “Population parameters are written in Greek” rule of thumb).

For each scenario please correctly identify the null hypothesis, denoted H_0 , and the alternative hypothesis, denoted H_A . The first two questions are multiple choice to give you the idea. Remember that all possible situations (eg the entire real number line) has to be in either the null or alternative hypothesis.

HINT 1: The “equal” part will always go with the null hypothesis in this class and does generally in the vast majority of modern research. The only time it doesn’t, to the best of my knowledge, is equivalency testing.

HINT 2: The $\hat{}$ (read “hat”) in statistics means “estimated”. So the estimated proportion is $\hat{\pi}$ for example.

1. A large company recently hired an advertising agency. The CEO of the company is interested in if more people recognize the company name now that the advertising is being ran from an outside company. The old proportion of people who recognized the company was .47. What is the correct Null Hypothesis and Alternative Hypothesis?

H_0

A. $\pi \leq .47$

B. $\hat{\pi} = .47$

C. $\pi = .47$

D. $\hat{\pi} = p$

H_A

A. $\hat{\pi} > .47$

B. $\pi > .47$

C. $\pi \neq .47$

D. $\pi \geq .47$

In this situation it’s assumed the CEO wants the company to have better name recognition than what they previously achieved. So here the alternative will be....

$$H_A : \pi > .47$$

Since the alternative is everything above .47, we want the null hypothesis to be everything else which in this case is equal to or below .47.

$$H_0 : \pi \leq .47$$

Note that we are using π here because we are talking about proportions of people who know the company name. Big hint is if you ever see the word “proportion” it means we are going to be using π most likely.

2. I happen to drink a lot of tea. About 6 cups in a day actually and occasionally some coffee to top it off (which is significantly down from when I originally wrote this problem in grad school which had it at 9 cups of tea and "commonly"). I get a lot of my tea down in the East Village of Des Moines at Gung Fu Tea where I generally buy tea in 4oz packages. Because of the amount of tea I buy from them I was once concerned they were, on average, underfilling the packages of tea. So over the course of a year I recorded the weight of tea in each 4oz package I bought on my own scales at home. After the year was over I found the mean weight of tea packages that I had bought. Please state the Null Hypothesis and Alternative Hypothesis. Note we are using μ for this problem because we are talking about averages/means.

H_0

- A. $\mu \geq 4$
- B. $\hat{\mu} = 4$
- C. $\mu = 4$
- D. $\hat{\mu} = \bar{y}$

H_A

- A. $\hat{\mu} < 4$
- B. $\mu < 4$
- C. $\mu \neq 4$
- D. $\mu \geq 4$

So here I'm interested if the bags are, on average, underfilled. I'm not going to say anything if the bags have the advertised weight (4oz) or higher weight (e.g. 5oz). So our null and alternative hypothesis then is....

$$H_0 : \mu \geq 4$$

$$H_A : \mu < 4$$

where μ is the average bag of tea weight.

NOTE: We traditionally write the null hypothesis first. I don't know why but that is the agreed upon system; I won't take off points if you write H_A first but be warned there will be people who think it "strange" (in the same way as "the red, big ball" is strange)

3. A business owner is interested to see if the a new manufacturer of sweatshirts is producing the sweatshirts at a better quality than the old manufacturer. To this end, the owner has a load of sweatshirts inspected to find the proportion that are defective. The old manufacturer had a defective proportion of about .04. Please state the Null Hypothesis and Alternative Hypothesis.

Here we are talking about proportions so we will be using π . Specifically, we are going to let π be the proportion of defective sweaters from the new manufacturer. Contextually the business owner wants to know if the sweaters are being produced at a better quality which corresponds to a lower proportion of defects. So....

$$H_0 : \pi \geq .04$$

$$H_A : \pi < .04$$

4. There is a professor at the University of Maine (Go Black Bears!!!) who is infamous for having his classes run over time. In a given semester an average UMaine professor would run over time with a proportion of about .10. Students in his class, after being taught hypothesis testing for proportions, recorded the proportion of days he ran over time and then performed a hypothesis test. Please state the Null Hypothesis and Alternative Hypothesis.

There was some confusion on this one and the next. The idea here is that the professor (my undergraduate advisor actually, he's awesome) who would regularly go over time. In this situation we are (with love) teasing him for it. So our goal isn't to show he lets classes go on time more than his peers, instead we want to show that, with statistical evidence, he let us out late more than other profs.

So we are working with proportions so we need to use π . And we are thinking that his peers let us students out late with a proportion of .10 so....

$$H_0 : \pi \leq .10$$

$$H_A : \pi > .10$$

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5. Let's tweak the above question: There is a professor at the University of Maine who is infamous for having his classes run over time. In a given semester an average UMaine professor would run over time by an average of about 1 minute. Students in his class, after being taught hypothesis testing for means, recorded the time he ran over through the course of the semester and then performed a hypothesis test. Please state the Null Hypothesis and Alternative Hypothesis.

Alright, let's tweak our answer. Instead of proportions we are talking about means so we will use μ instead of π . Here, we will say μ is the average time the professor let us out (so $\mu = 5$ would imply he let us out late by five minutes on average). The starting point would be that he acts like other professors and would let us out on average late by 1 minute. We are again trying to prod him some with statistical evidence that he lets us out later than his peers.

$$H_0 : \mu \leq 1$$

$$H_A : \mu > 1$$

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6. A teacher in a middle school is interested to see if a new method of teaching reading would improve the literacy rate of their students. So for one year they taught with the new method and then looked at the standardized test scores for their students. From past standardized testing, the teacher knew that around 82% of their students would pass. Please state the Null Hypothesis and Alternative Hypothesis.

So it doesn't really talk about proportions but it does talk about percentages (82%). Since a percentage is just a proportion multiplied by 100 we can work with proportions (e.g. .82).

The proportion that we are interested in, π is the proportion of students who pass their standardized test for literacy. Contextually we would assume that a teacher would embrace a new teaching method because it would improve literacy (as an educator we really really don't want our students to become less literate). So we are hoping the new method increases the proportion of students who pass which was .82.

$$H_0 : \pi \leq .82$$

$$H_A : \pi > .82$$

7. Cyride (bus system in Ames IA) tries to be on time as much as possible and to this end management established an aggressive new policy. After the policy went into effect, management wanted to know if the busses were delayed because too many bus stops had people running to catch the bus. To this end the drivers were instructed to keep a tally for how many bus stops they had to wait for “runners”. The old scheduling method assumed the busses would wait at a rate of 1 in 5 bus stops. Please state the Null Hypothesis and Alternative Hypothesis. NOTE: This is a made up scenario as CyRide bus drivers would never wait. At times they would make eye contact with you as they drove away.

So, originally the busses would be delay at 1 in 5 bus stops which works out to be $\frac{1}{5} = .20$. So from there we can see we will be working with proportions and π .

The management implemented an aggressive new policy which seems to imply they want the proportion of delayed busses to decrease.

$$H_0 : \pi \geq .20$$

$$H_A : \pi < .20$$
